

Database Systems - MCQ Question Bank (Set 3)

20 Multiple Choice Questions | Includes 5 repeated from earlier sets | Answer Key on Last Page

1. In the relational model, a table is also referred to as a:

- A. Tuple
- B. Attribute
- C. Relation
- D. Domain

2. A relation is in First Normal Form (1NF) if:

- A. It has no transitive dependency
- B. All attributes contain only atomic (indivisible) values
- C. Every determinant is a candidate key
- D. It has no foreign keys

3. Which of the following best describes a deadlock in a DBMS?

- A. A query that returns no results
- B. Two or more transactions waiting indefinitely for each other to release locks
- C. A failure of the database server hardware
- D. An error caused by invalid SQL syntax

4. What does the term 'referential integrity' ensure in a relational database?

- A. All tables have the same number of columns
- B. Foreign key values match a primary key value in the referenced table, or are NULL
- C. Data is encrypted during transmission
- D. Queries execute within a fixed time limit

5. A schedule is said to be serializable if it is equivalent to some:

- A. Concurrent schedule
- B. Serial schedule
- C. Partial schedule
- D. Deadlocked schedule

6. Which of the following best defines data independence in a DBMS?

- A. The ability to modify schema at one level without affecting schema at another level
- B. Storing data on multiple servers
- C. Allowing multiple users to access data simultaneously
- D. Encrypting data at rest

7. Which SQL constraint ensures that a column cannot store duplicate values?

- A. CHECK
- B. UNIQUE
- C. DEFAULT
- D. NOT NULL

8. In query optimization, the term 'query plan' refers to:

- A. A backup strategy for the database
- B. A sequence of steps the DBMS uses to execute a query efficiently

- C. A diagram of the database schema
- D. A user access control policy

9. Which of the following is true about a self-join?

- A. It joins a table with a different table
- B. It joins a table to itself using aliases
- C. It can only be used with the DELETE statement
- D. It removes duplicate rows automatically

10. Which recovery technique uses a log of before and after images of data to undo/redo transactions?

- A. Shadow paging
- B. Write-ahead logging
- C. Checkpointing only
- D. Index scanning

11. Multivalued dependency is primarily addressed in which normal form?

- A. 2NF
- B. 3NF
- C. BCNF
- D. Fourth Normal Form (4NF)

12. Which of the following best describes a trigger in a database?

- A. A query that runs only once at database creation
- B. A stored procedure that automatically executes in response to certain events on a table
- C. A type of primary key
- D. A method to compress data

13. Which SQL keyword is used to combine the results of two SELECT statements, removing duplicates?

- A. JOIN
- B. UNION
- C. UNION ALL
- D. MERGE

14. What is the primary role of a database schema?

- A. To store actual data values
- B. To define the logical structure of the database
- C. To encrypt sensitive fields
- D. To manage user login credentials

15. Which of the following best describes optimistic concurrency control?

- A. Transactions acquire locks before accessing data
- B. Transactions proceed without locking and are validated before commit
- C. Only one transaction is allowed at a time
- D. Transactions are always executed serially

16. In a star schema used for data warehousing, the central table is called the:

- A. Dimension table
- B. Fact table

- C. Lookup table
- D. Staging table

17. Which of the following is NOT a valid SQL data type category?

- A. Numeric
- B. Character/String
- C. Date/Time
- D. Executable

18. Which term describes the property that a completed transaction's changes persist even after a system failure?

- A. Atomicity
- B. Isolation
- C. Durability
- D. Consistency

19. Which of the following best describes denormalization?

- A. Splitting tables to remove redundancy
- B. Intentionally introducing redundancy to improve read performance
- C. Encrypting normalized tables
- D. Removing all foreign keys from a schema

20. Which SQL command is used to grant a user access privileges to a database object?

- A. GRANT
- B. ALLOW
- C. PERMIT
- D. ACCESS

Answer Key

1. C 2. B 3. B 4. B

5. B 6. A 7. B 8. B

9. B 10. B 11. D 12. B

13. B 14. B 15. B 16. B

17. D 18. C 19. B 20. A